

# Wetland Placement Pipeline

## Overview

This pipeline places dynamic wetland cells in a river network by:

1. Detecting wetland area jumps (significant increases in accumulated WAD2M)
2. Placing wetlands at tributary confluences
3. Spacing wetlands along inter-node segments by flow distance
4. Extracting local wetland extent time series

## Key datasets:

- WAD2M wetland inundation (2000-2018)
- River network connectivity (from LDD)
- Upstream drainage area
- Flow-accumulated wetland area

## Pipeline Structure

| Script                           | Purpose                           |
|----------------------------------|-----------------------------------|
| 0_wetlands_network_build.py      | Build network topology from LDD   |
| 1_wetlands_wad2m_accumulation.py | Accumulate WAD2M upstream         |
| 2_wetlands_placing.py            | Place wetlands via jump detection |
| 3_wetland_extent.py              | Extract local wetland extents     |

Run sequentially:  $0 \rightarrow 1 \rightarrow 2 \rightarrow 3$

## Folder Structure

### Input Data (1\_data/)

| File                           | Description                      |
|--------------------------------|----------------------------------|
| ldd_{case}.map                 | Flow direction (LDD format)      |
| ups_{case}.map                 | Upstream area [km <sup>2</sup> ] |
| cellarea_{case}.map            | Cell area [m <sup>2</sup> ]      |
| WAD2M_._{case}.nc              | Wetland fraction [0-1]           |
| upstream_connect_{case}.pkl    | Direct upstream links            |
| allupstream_connect_{case}.pkl | All upstream cells               |

### Results (2\_results/)

#### Script 0 - Network:

| File                     | Description |
|--------------------------|-------------|
| 0_network_validation.png | QC plot     |

#### **Script 1 - Accumulation:**

| File                                | Description             |
|-------------------------------------|-------------------------|
| 1_WAD2M_upstream_area_km2_{case}.nc | Monthly upstream area   |
| 1_WAD2M_validation_{case}.png       | QC plot                 |
| 1_WAD2M_max_accumulated_{case}.tif  | Max accumulated (QGIS)  |
| 1_WAD2M_max_extent_{case}.tif       | Max local extent (QGIS) |

#### **Script 2 - Placement:**

| File                               | Description        |
|------------------------------------|--------------------|
| 2_placed_wetlands_{case}.csv       | Placed wetlands    |
| 2_node_pairs_{case}.csv            | Node connections   |
| 2_removed_nodes_{case}.csv         | Filtered nodes     |
| 2_wetland_nodes_map_{case}.png     | Nodes map          |
| 2_wetland_placement_map_{case}.png | Placement map      |
| 2_river_mask_{case}.tif            | River mask (QGIS)  |
| 2_wetland_nodes_{case}.shp         | Nodes shapefile    |
| 2_placed_wetlands_{case}.shp       | Wetlands shapefile |

#### **Script 3 - Extent:**

| File                                  | Description          |
|---------------------------------------|----------------------|
| 3_wetland_extent_km2_total_{case}.csv | Monthly total extent |
| 3_wetland_extent_km2_local_{case}.csv | Monthly local extent |
| 3_.._daily_lininterp_{case}.csv       | Daily (interpolated) |
| 3_.._daily_flat_{case}.csv            | Daily (flat)         |
| 3_wetland_metadata_{case}.csv         | Wetland metadata     |
| 3_placed_wetlands_filtered_{case}.csv | Final wetlands       |
| 3_removed_wetlands_{case}.csv         | Removed wetlands     |
| 3_placed_wetlands_filtered_{case}.shp | Final shapefile      |

## Script 0: Network Building

**Purpose:** Build river network topology from LDD.

### Inputs

| File           | Description              |
|----------------|--------------------------|
| ldd_{case}.map | Flow direction (LDD 1-9) |
| ups_{case}.map | Upstream area (for QC)   |

### Configuration

| Parameter | Description            |
|-----------|------------------------|
| CASE_NAME | Case ID (e.g., “juba”) |

### Outputs

| File                           | Description           |
|--------------------------------|-----------------------|
| upstream_connect_{case}.pkl    | Direct upstream links |
| allupstream_connect_{case}.pkl | All upstream cells    |
| 0_network_validation.png       | QC plot               |

## Script 1: WAD2M Accumulation

**Purpose:** Compute upstream wetland area by routing WAD2M along network.

### Inputs

| File                           | Description                 |
|--------------------------------|-----------------------------|
| upstream_connect_{case}.pkl    | From Script 0               |
| allupstream_connect_{case}.pkl | From Script 0               |
| ldd_{case}.map                 | Flow direction              |
| ups_{case}.map                 | Upstream area               |
| cellarea_{case}.map            | Cell area [m <sup>2</sup> ] |
| WAD2M_.__{case}.nc             | Wetland fraction            |

### Configuration

| Parameter     | Default | Description                |
|---------------|---------|----------------------------|
| CASE_NAME     | “juba”  | Case ID                    |
| UPS_THRESHOLD | 0       | Min upstream cells (0=all) |

## Outputs

| File                                | Description           |
|-------------------------------------|-----------------------|
| 1_WAD2M_upstream_area_km2_{case}.nc | Monthly upstream area |
| 1_WAD2M_validation_{case}.png       | QC plot               |
| 1_WAD2M_max_accumulated_{case}.tif  | Max accumulated       |
| 1_WAD2M_max_extent_{case}.tif       | Max local extent      |

## Script 2: Wetland Placement

**Purpose:** Detect jump nodes and place wetlands at confluences and along segments.

### Inputs

| File                                | Description      |
|-------------------------------------|------------------|
| WAD2M_.._{case}.nc                  | Wetland fraction |
| ups_{case}.map                      | Upstream area    |
| ldd_{case}.map                      | Flow direction   |
| upstream_connect_{case}.pkl         | Upstream links   |
| 1_WAD2M_upstream_area_km2_{case}.nc | Accumulated area |

### Configuration

| Parameter            | Default | Description                         |
|----------------------|---------|-------------------------------------|
| CASE_NAME            | “juba”  | Case ID                             |
| UPS_THRESHOLD        | 15000   | Min river area [km <sup>2</sup> ]   |
| JUMP_THRESHOLD       | 2500    | Min jump to flag [km <sup>2</sup> ] |
| SHORT_DIST_KM        | 200     | 0 wetlands if shorter               |
| MEDIUM_DIST_KM       | 550     | 1 wetland if shorter                |
| LONG_DIST_KM         | 900     | 2 wetlands if shorter               |
| VLONG_DIST_KM        | 1250    | 3 wetlands; 4 if longer             |
| MIN_NODE_DISTANCE_KM | 50      | Min node spacing [km]               |

### Outputs

| File                         | Description      |
|------------------------------|------------------|
| 2_placed_wetlands_{case}.csv | Placed wetlands  |
| 2_node_pairs_{case}.csv      | Node connections |

| File                               | Description        |
|------------------------------------|--------------------|
| 2_removed_nodes_{case}.csv         | Filtered nodes     |
| 2_wetland_nodes_map_{case}.png     | Nodes map          |
| 2_wetland_placement_map_{case}.png | Placement map      |
| 2_river_mask_{case}.tif            | River mask         |
| 2_wetland_nodes_{case}.shp         | Nodes shapefile    |
| 2_placed_wetlands_{case}.shp       | Wetlands shapefile |

### Script 3: Extent Extraction

**Purpose:** Extract monthly climatology and calculate local wetland contribution.

#### Inputs

| File                                | Description           |
|-------------------------------------|-----------------------|
| 2_placed_wetlands_{case}.csv        | Wetland locations     |
| 1_WAD2M_upstream_area_km2_{case}.nc | Accumulated area      |
| allupstream_connect_{case}.pkl      | Upstream connectivity |

#### Configuration

| Parameter                  | Default | Description                           |
|----------------------------|---------|---------------------------------------|
| CASE_NAME                  | “juba”  | Case ID                               |
| MIN_LOCAL_EXTENT_KM2       | 250     | Min extent to keep [km <sup>2</sup> ] |
| INCLUDE_INTERNODE_WETLANDS | True    | Include inter-node?                   |

#### Outputs

| File                                  | Description        |
|---------------------------------------|--------------------|
| 3_wetland_extent_km2_total_{case}.csv | Monthly total      |
| 3_wetland_extent_km2_local_{case}.csv | Monthly local      |
| 3_.._daily_lininterp_{case}.csv       | Daily interpolated |
| 3_.._daily_flat_{case}.csv            | Daily flat         |
| 3_wetland_metadata_{case}.csv         | Metadata           |
| 3_placed_wetlands_filtered_{case}.csv | Final wetlands     |
| 3_removed_wetlands_{case}.csv         | Removed wetlands   |
| 3_placed_wetlands_filtered_{case}.shp | Final shapefile    |

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## Dependencies

- numpy
- xarray
- pandas
- netCDF4
- rasterio
- rioxarray
- matplotlib
- geopandas
- shapely
- plotly
- helper\_functions (local)